ACM 104 HW #2

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1. Supspaces

a) Not a subspace

Singular matrices arent closed under addition.

Counterexample: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = I$ which is invertible, but both matrices are singular.

b) subspace

$tr(A+B) = tr(A) + tr(B)$

$tr(cA) = c \cdot tr(A)$

$\therefore$ closed under addition and multiplication and contain 0

c) not a subspace

The zero function isn't in W since $(0 \cdot 0 \neq 1)$.

d) subspace

zero: $z(1/2) = 0$; $\int_0^1 z(t)\,dt = \int_0^1 0\,dt = 0$

closure(+): $h(x) = f(x) + g(x) = \int_0^1 f(t)dt + \int_0^1 g(t)dt$

$\int_0^1 h(t)\,dt = \int_0^1 (f(t) + g(t))dt = \int_0^1 f(t)dt + \int_0^1 g(t)dt$

$\therefore f+g \in W$

closure(·): $f(1/2) = \int_0^1 f(t)\,dt$; $h(x) = c f(x)$

$\int_0^1 h(t)dt = \int_0^1 c f(t)dt = c\int_0^1 f(t)dt$

$\therefore h(1/2) = \int_0^1 h(t)dt$, so $cf \in W$

e) subspace

zero: $\nabla \cdot 0 = \frac{\partial 0}{\partial x} + \frac{\partial 0}{\partial y}$

closure(+): $\nabla \cdot (u+v) = \left(\frac{\partial(u_1+v_1)}{\partial x} + \frac{\partial(u_2+v_2)}{\partial y}\right)$

$= \left(\frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y}\right) + \left(\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y}\right)$

$= \nabla \cdot u + \nabla \cdot v = 0 + 0 = 0$

$\therefore u+v \in W$

closure $(\cdot)$: $\nabla \cdot (cv) = \frac{\partial (cv_1)}{\partial x} + \frac{\partial (cv_2)}{\partial y}$

$$= c\left(\frac{\partial v_1}{\partial x}\right) + c\left(\frac{\partial v_2}{\partial y}\right)$$

$$= c\left(\frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y}\right) = c(\nabla \cdot v) = c \cdot 0 = 0$$

$\therefore cv \in W$

f) not a subspace

Counterexample: Let V_1 be x-axis and V_2 be y-axis.
Both are subspaces.

Let $u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $u \in V_1$, so $u \in V_1 \cup V_2$

Let $v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $v \in V_2$, so $v \in V_1 \cup V_2$

$u + v = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin V_1 \cup V_2$ $\therefore$ not closed

g) subspace

zero: Since V_1, V_2 are subspaces, they both must contain 0 vector.

$0 \in V_1 \cap V_2$

closure $(+)$: $u, v \in V_1 \cap V_2$

- This implies $u, v \in V_1$, since V_1 is a subspace.
- Also implies $u, v \in V_2$, since V_2 is a subspace.
- Since $u + v \in V_1, V_2$; $u + v \in V_1 \cap V_2$

closure $(\cdot)$: Let $v \in V_1 \cap V_2$

- This implies $v \in V_1$. Since V_1 is a subspace, $cv \in V_1$.
- Also implies $v \in V_2$. Since V_2 is a subspace, $u + v \in V_2$
- Since $cv \in V_1, V_2$; $cv \in V_1 \cap V_2$.

2. Polynomials and Exponentials

a) Yes linearly independent

$$c_1 p_1(x) + c_2 p_2(x) + c_3 p_3(x) = 0$$

$$c_1(x^2-3) + c_2(2-x) + c_3(x+1)^2 = 0$$

$$(c_1+c_3)x^2 + (-c_2+2c_3)x + (-3c_1+2c_2+c_3) = 0$$

$c_1 + c_3 = 0$; $-c_2 + 2c_3 = 0$; $-3c_1 + 2c_2 + c_3 = 0$

$-3(-c_3) + 2(2c_3) + c_3 = 0$, $8c_3 = 0$; $c_3 = 0$, $c_1 = 0$, $c_2 = 0$

Only trivial solution ∴ linearly independent

b) $\dim P^{(2)} = 3$. 3 lin. indep. vectors in a 3D space span it

Yes, they span $P(2)$

c) $c_1 p_1 + c_2 p_2 + c_3 p_3 = 1$

$c_1 + c_3 = 0$; $-c_2 + 2c_3 = 0$; $-3c_1 + 2c_2 + c_3 = 1$

$8c_3 = 1$ ∴ $c_3 = 1/8$, $c_1 = -1/8$, $c_2 = 1/4$

coordinates of $q(x) = 1$ in basis $\{p_1, p_2, p_3\}$ are $(-\frac{1}{8}, \frac{1}{4}, \frac{1}{8})$

d) yes, they are linearly independent

$$c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) = 0$$

$$c_1 e^x + c_2 x e^x + c_3 x^2 e^x = 0$$

$$e^x(c_1 + c_2 x + c_3 x^2) = 0$$

$$c_1 + c_2 x + c_3 x^2 = 0$$

must hold for all x

polynomial is zero for all x iff its coefficients are zero

∴ $c_1 = 0$, $c_2 = 0$, $c_3 = 0$ → only solutions

∴ f_1, f_2, f_3 are lin. indep.

3. Fibonacci Sequences

a) zero vector: $x_1 = 0,\ x_2 = 0 \rightarrow x_3 = x_2 + x_1 = 0 + 0 = 0$

$x_4 = x_3 + x_2 = 0 + 0 = 0$. By induction, $x_n = 0$ for all n.

closed (+): $f_1 = (x_1, x_2, \ldots)$; $f_2 = (y_1, y_2, \ldots) \in \mathcal{F}$

$x_n = x_{n-1} + x_{n-2}$ and $y_n = y_{n-1} + y_{n-2}$ for all $n \geq 3$

$f_1 + f_2 = (x_1 + y_1, x_2 + y_2, \ldots)$.

Let $g = (z_1, z_2, \ldots)$ where $z_n = x_n + y_n$. Check if g works for $n \geq 3$:

$z_n = x_n + y_n$

$z_n = (x_{n-1} + x_{n-2}) + (y_{n-1} + y_{n-2})$

$z_n = (x_{n-1} + y_{n-1}) + (x_{n-2} + y_{n-2})$

$z_n = z_{n-1} + z_{n-2}$ $\therefore$ $f_1 + f_2 \in \mathcal{F}$

closed($\cdot$): $f = (x_1, x_2, \ldots) \rightarrow x_n = x_{n-1} + x_{n-3}$ for $n \geq 3$

$\alpha f = (\alpha x_1, \alpha x_2, \ldots)$. Let $h = (w_1, w_2, \ldots)$

$w_n = \alpha x_n$

$w_n = \alpha(x_{n-1} + x_{n-2})$

$w_n = \alpha x_{n-1} + \alpha x_{n-2}$

$w_n = w_{n-1} + w_{n-2}$ $\therefore$ $\alpha f \in \mathcal{F}$

b) $\dim(\mathcal{F}) = 2$ b/c it's uniquely determined by first two terms: x_1 / x_2 (rest determined by recurrence).

We can build basis by choosing two independent sequences.

- Let $b_1 = (1, 0, 1, 1, 2, 3, 5)$
- Let $b_2 = (0, 1, 1, 2, 3, 5, 8)$

B spans $\mathcal{F}$: Let $f = (x_1, x_2, \ldots) \in \mathcal{F}$

We want to find $c_1, c_2 \ni f = c_1 b_1 + c_2 b_2$

$$f = c_1(1, 0, 1, \ldots) + c_2(0, 1, 1, \ldots)$$
$$= (c_1, c_2, c_1 + c_2, \ldots)$$

- First term: $x_1 = c_1$
- Second term: $x_2 = c_2$

So any $f \in \mathcal{F}$ can be written as $f = x_1 b_1 + x_2 b_2$

Since $x_1 b_1$ and $x_2 b_2 \in \mathcal{F}$ (by closure of $\cdot$), their sum is also in $\mathcal{F}$ (by closure of $+$). $\therefore$ B spans $\mathcal{F}$

B is lin. indep.

$$c_1 b_1 + c_2 b_2 = 0$$
$$c_1(1, 0, 1, \ldots) + c_2(0, 1, 1, \ldots) = (0, 0, 0, \ldots)$$
$$(c_1, c_2, c_1 + c_2, \ldots) = (0, 0, 0, \ldots)$$

Only trivial solution $\therefore$ lin. indep.

B is lin indep and spans $\mathcal{F}$ so basis:

$$B = \{(1, 0, 1, 1, 2, \ldots), (0, 1, 1, 2, 3, \ldots)\}$$

c) From (b), for any sequence $f = (x_1, x_2, \ldots)$, its coordinates in basis $B = \{b_1, b_2\}$ are (x_1, x_2)

For original Fibonacci, $x_1 = 1$, $x_2 = 1$

$\therefore$ Coodinates in basis are $(1, 1)$.

4. Fundemental Matrix Subspaces

a) The two cols $\begin{bmatrix} 2 \\ 2 \\ 20 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 2 \\ 25 \end{bmatrix}$ are linearly independent, so rank $A = 2$.

- dim Ker $A = 2 -$ rank $A = 0$
- dim im $A =$ rank $A = 2$
- dim cok $A = 3 -$ rank $A = 1$
- dim coim $A = 2 -$ dim Ker $A = 2$

b) Kernel: Ker $A = \{0\}$ (Basis empty set) $\emptyset$

Image: $\left\{ \begin{bmatrix} 2 \\ 2 \\ 20 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 25 \end{bmatrix} \right\}$

Coimage: $\left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right\}$ (row echelon form of A)

Kernel: Solving $Ax = 0$. From REF, $2x_1 = 0$, $2x_2 = 0$, $x_1 = 0$, $x_2 = 0$
only trivial solution $x = 0$

Basis$(N(A)) = \emptyset$

Cokernel: Solve $A^T z = 0$

$$A^T z = \begin{bmatrix} 2 & 2 & 20 \\ 0 & 2 & 25 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

z_3 is free variable. Let $z_3 = t$

$2z_2 + 25z_3 = 0 \rightarrow z_2 = -\frac{25}{2}t$

$2z_1 + 2z_2 + 20z_3 = 0 \rightarrow 2z_1 - 25t + 20t = 0 \rightarrow z_1 = \frac{5}{2}t$

$z = t \begin{bmatrix} 5/2 \\ -25/2 \\ 1 \end{bmatrix} \rightarrow$ pick $t = 2 \rightarrow$ Basis $= \left\{ \begin{bmatrix} 5 \\ -25 \\ 2 \end{bmatrix} \right\}$

c) Each col j can be written as:

$$\text{Col } j = \begin{bmatrix} j \\ n+j \\ 2n+j \\ \vdots \\ (n-1)n+j \end{bmatrix} = j \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + n \begin{bmatrix} 0 \\ 1 \\ 2 \\ \vdots \\ n-1 \end{bmatrix}$$

Therefore all cols lie in span of two vectors so $\text{rank}(B)=2$

Image:

$$\text{Basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ \vdots \\ n-1 \end{bmatrix} \right\}$$

Kernel: We need $Bx=0$, requiring $\sum_{j=1}^{n} x_j = 0$ and $\sum_{j=1}^{n} j \cdot x_j = 0$

For $j = 3, 4, \ldots, n$, $v_j = [j-2, -(j-1), 0, \ldots, 1, \ldots, 0]^T \to$ basis

$$\text{Basis for } \ker(B): \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \ldots, \begin{bmatrix} n-2 \\ -(n-1) \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \right\}$$

Coimage: Non-zero RREF rows

$$\text{Basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ \vdots \\ n-1 \end{bmatrix} \right\}$$

Cokernel: Sum from above: $(k-2)-(k-1)+0+\ldots+1+\ldots+0=0$ ✓

Weighted Sum: $0 \cdot (k-2) + 1 \cdot (-(k-1)) + \ldots + (k-1) \cdot 1 = 0$ ✓

$$\text{Basis}: \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \text{im}(B), \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \text{im}(B), \ldots, \begin{bmatrix} n-2 \\ -(n-1) \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} + \text{im}(B) \right\}$$

ACM/IDS 104 - Problem Set 2 - MATLAB Problems

Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

Problem 5 (10 points) Fundamental Matrix Subspaces

Your task for this problem is to write a function that takes a matrix `A` as its argument, and outputs four matrices: `K, I, cK` and `cI` where:

- Columns of `K` form a basis of the kernel of `A`. If $\ker A = \{0\}$, then `K` must be a zero vector of the appropriate dimension.
- Columns of `I` form a basis of the image of `A`. If $\mathrm{im} A = \{0\}$, then `I` must be a zero vector of the appropriate dimension.
- Columns of `cK` form a basis of the cokernel of `A`. If $\mathrm{coker} A = \{0\}$, then `cK` must be a zero vector of the appropriate dimension.
- Columns of `cI` form a basis of the coimage of `A`. If $\mathrm{coim} A = \{0\}$, then `cI` must be a zero vector of the appropriate dimension.

Move to the bottom of this livescript to write the function.

Now, let us test our function:

```
A = magic(6); % feel free to define A as you like
[K, I, cK, cI] = subspacer(A); % this is how you call a MATLAB function
disp(K);
```

```
   -0.4714
   -0.4714
    0.2357
    0.4714
    0.4714
   -0.2357
```

```
disp(I);
```

```
   -0.4082    0.5574    0.0456   -0.4182    0.3092
   -0.4082   -0.2312    0.6301   -0.2571   -0.5627
   -0.4082    0.4362    0.2696    0.5391    0.1725
   -0.4082   -0.3954   -0.2422   -0.4590    0.3971
   -0.4082    0.1496   -0.6849    0.0969   -0.5766
   -0.4082   -0.5166   -0.0182    0.4983    0.2604
```

```
disp(cK);
```

```
    0.5000
   -0.0000
   -0.5000
   -0.5000
```

```
    0.0000
    0.5000
```

```
disp(cI);
```

```
   -0.4082    0.6234   -0.3116    0.2495   -0.2511
   -0.4082   -0.6282    0.3425    0.1753   -0.2617
   -0.4082   -0.4014   -0.7732   -0.0621    0.1225
   -0.4082    0.1498    0.2262   -0.4510   -0.5780
   -0.4082    0.1163    0.2996    0.6340    0.3255
   -0.4082    0.1401    0.2166   -0.5457    0.6430
```

START HERE by writing the function:

```
function [K, I, cK, cI] = subspacer(A)
%{
This is the MATLAB function syntax.
-> [K, I, cK, cI] are the outputs of the function.
-> "subspacer" is the name of the function. (you can change that if
                            you wish but make sure you change
                            every function call as well!)
-> A is the argument of the function.
%}
[m, n] = size(A);
r = rank (A);
%{
We start by finding out the dimensions and rank of A.
Let us consider the matrix K. There exist 2 cases:
1) The kernel is trivial i.e. kerA = {0}
2) The kernel is not trivial -> Hint: use null()
Complete the following if/else statement.
%}
if r == n % this condition is done for you
    K = zeros(n,1);
else
    K = null(A);
end
%{
Now, let us consider the matrix cK.
As above, there exist 2 cases. Remember, you can use ' to
transpose a matrix.
Write a similar if/else statement to produce cK.
%}
if r == m
    cK = zeros(m,1);
else
    cK = null(A');
end
```

```
%{
For the image I and coimage cI, there exists only 1 condition
we must test, and that is if rankA = 0. With this in mind,
complete the following if/else statement.
-> Hint: orth() is useful here.
%}
if r == 0
    I = zeros(m,1);
    cI = zeros(n,1);
else
    I = orth(A);
    cI = orth(A');
end
end
```